AR 2EB

ALAMEDA RESEARCH
2: ELECTRIC BOOGALOO
LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

Joby Aviation

NYSE : JOBY · Young-Company DCF (Damodaran)

16 May 2026 · the qualitative

$10.50

$10.3B mkt cap · 984M sh · $2.5B cash, zero debt

THE CENTRAL QUESTION

Will Joby capture meaningful share of a $250B global UAM TAM by 2036 — or is today's $10B market cap pricing certainty that competitive eVTOL economics don't support?

Joby trades at ~$10.3B against $24M of Q1 2026 revenue and $700M annualized cash burn. The $2.5B cash position (post Feb 2026 $1.2B raise) provides ~3 years of runway. FAA Type Certification is in Stage 4 (final), Dubai exclusive operations begin in 2026, and the Toyota manufacturing partnership unlocks scale. Pre-revenue category-defining companies present a Damodaran problem: the standard 5-year DCF generates nonsense. The young-company framework asks what mature TAM share is plausible, what terminal margins look like at scale, and what probability of outright failure. Three scenarios, weighted; show your work.

PROBABILITY-WEIGHTED EXPECTED VALUE

$8.44 expected fair value today

-19.6% vs spot $10.50

FORWARD COMPOUNDED VALUE

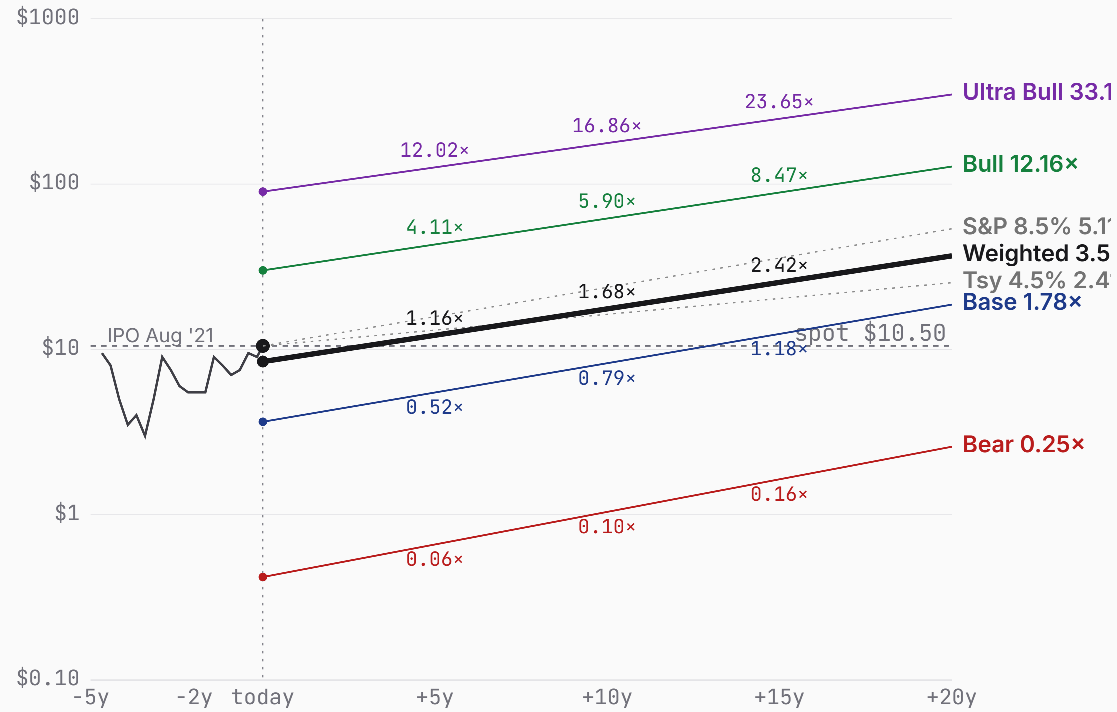
+5y	+10y	+15y	+20y
$12.18	$17.60	$25.44	$36.80
1.16× spot	1.68× spot	2.42× spot	3.51× spot

WEIGHTING RATIONALE

Bear 30% Aerospace startups historically fail at high rates; SR3 timing slipped industry-wide.
Base 55% Modal outcome: Joby + Archer split US with aerospace-blend margins.
Bull 12% Compound conditional chain; joint probability lands at 10-15% despite individually plausible pieces.
Ultra Bull 3% Tail of tails: dominant global UAM + early autonomous pivot + software-take-rate multiple expansion.

Bull (12%) + Ultra Bull (3%) together contribute $6.31 — 75% of the $8.44 expected value despite 15% combined probability. Today's spot ($10.50) sits 24% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED



BEAR $0.42	**BASE** $3.65	**BULL** $30.05	**ULTRA BULL** $90.01
-96.0% vs spot	-65.2% vs spot	+186.2% vs spot	+757.2% vs spot
TAM 0.6% P(fail) 35%	TAM 2.8% P(fail) 18%	TAM 8.0% P(fail) 8%	TAM 16.0% P(fail) 4%
S2C 0.8 Prob 30%	S2C 1.2 Prob 55%	S2C 1.6 Prob 12%	S2C 1.8 Prob 3%
Cert delays; mfg stumbles; China wins.	Joby + Archer split US; 13% margin.	Western eVTOL leader; SaaS margins.	Global UAM dominance; SaaS-take-rate.

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Joby Aviation scenario narratives

PROBABILITY WEIGHTS

Bear 30% Base 55% Bull 12% Ultra Bull 3% · Weighted expected $8.44 (-19.6% vs spot)

BEAR $0.42

-96.0% vs spot

Probability 30%

Cert delays; mfg stumbles; China wins.

WHY 30%

Aerospace startups historically fail at high rates even with funding. SR3 audit timing has slipped industry-wide; Stage 4 final testing routinely extends 12-24 months beyond projections. Manufacturing scaling at Dayton is the hardest unproven part — Tesla took years on Model 3 with mature automotive suppliers. The 30% weight acknowledges that the $2.5B cash + Toyota manufacturing partnership materially reduce execution risk relative to a typical aerospace startup, but eVTOL unit economics remain fundamentally unproven at fleet scale. Even with strong execution, the category itself could fail to generate profitable demand. Within-scenario 35% p_fail captures Joby-specific bankruptcy; the 30% scenario weight captures the broader 'bear world' where eVTOL economics don't scale even if Joby executes.

WHAT HAPPENS

The pessimistic case is brutal for eVTOL economics. FAA Type Certification slips from 2026 into 2027 or later — the SR3 audit is complete but Stage 4 final testing has historically taken longer than companies project. The Dubai launch is real but on a single route; volume builds slowly. Manufacturing in Dayton scales below the 4-aircraft/month target.

Cash burn continues at $700M+ annualized. The $2.5B cash position lasts ~3 years, but commercial scaling requires far more capital. By FY30 Joby has raised another $2-3B at progressively lower prices, diluting holders 50%+. Aerospace economics don't reach software-like margins — at scale operating margin tops out around 5%, similar to legacy regional aviation operators.

BASE $3.65

-65.2% vs spot

Probability 55%

Joby + Archer split US; 13% margin.

WHY 55%

Modal outcome given de-risking factors already in place. FAA Type Cert progresses on schedule or slightly delayed; Toyota manufacturing scales incrementally rather than to full 500-aircraft/year; Dubai launches commercially but stays limited in scope; Joby and Archer effectively split the US market based on funding/partnership advantages. The 55% weight reflects that the base case requires the LEAST faith — no breakthrough business model, no dominant competitive position, no breakthrough regulatory cooperation. Just sequential execution on plans largely in motion. Result: real revenue with aerospace-blend operating margins (~13%, peer-set median between regional aviation at 8% and ADAS software at 30%), well below the software-economics framing implicit in today's market cap. Joby reaches commercial scale; the open question is at what margin.

WHAT HAPPENS

The middle path. FAA Type Certification comes through in late 2026 / early 2027. Joby's Dubai exclusivity through 2032 anchors Middle East revenue, US operations via the eIPP program scale through 2027-28, and the Toyota partnership delivers manufacturing scale at the Dayton facility. By FY36, ~1,400 aircraft and $7B revenue at 13% operating margin.

But the market splinters. Archer is the credible #2 in the US. EHang dominates China. Joby captures 2.8% of $250B global UAM TAM — meaningful but not dominant. Mature 13% operating margin reflects aerospace economics: better than legacy carriers (5-8%) but well below software (25%+). This is what the Damodaran framework produces with defensible peer-set anchors.

BULL $30.05

+186.2% vs spot

Probability 12%

Western eVTOL leader; SaaS margins.

WHY 12%

The bull case requires a chain of conditional events: FAA Type Cert on schedule by end-2026 AND Toyota Dayton scales to full 500-aircraft/year capacity by 2028-29 AND Dubai exclusivity template replicates in 3+ countries (Saudi Arabia, India, SE Asia) AND competition narrows (Archer ships meaningfully lower volumes, EHang stays regional to China) AND autonomous flight matures within 5-7 years AND Joby successfully pivots to software-take-rate model on partner aircraft. Even granting 60-70% probability per condition with correlation (i.e., if cert works Toyota probably scales), joint probability lands closer to 10-15%. Conjunctive probability is one of the most reliably underestimated quantities in forecasting — the bull case isn't blocked by any single piece looking unlikely, but by compound dependency. 15% reflects that this is a real possibility, just not central.

WHAT HAPPENS

The bull case is that Joby becomes the dominant Western eVTOL operator and manufacturer. FAA Type Certification on schedule in 2026 unlocks rapid scaling. The Toyota manufacturing partnership delivers the Dayton facility at full 500-aircraft/year capacity by FY28-29. Dubai exclusivity becomes a template — Joby wins similar long-term operating contracts in Saudi Arabia, India, and Southeast Asia.

Competition narrows. Archer ships but at lower volumes. EHang remains regional to China. By FY36 Joby operates 4,000 aircraft, generates $20B revenue, and captures 8% of $250B global UAM TAM. As autonomous flight capability matures (5-7 years away), Joby pivots to a software-take-rate model on partner-operated aircraft, reaching software-like margins. Mature operating margin reaches 28%.

ULTRA BULL $90.01

+757.2% vs spot

Probability 3%

Global UAM dominance; SaaS-take-rate.

WHY 3%

Tail of tails: requires every Bull condition to hit AND multiple expansion AND replicate internationally AND early autonomous pivot. Each individually 60-75% probable; joint at 3-5%. Genuine tail captures the scenario where Joby joins the software-aerospace-hybrid giants club.

WHAT HAPPENS

The ultra-bull case is that Joby becomes the dominant global UAM platform — not just Western leader. FAA cert on schedule + Toyota Dayton scales + Dubai template replicates in 5+ countries + autonomous flight matures within 5-7 years + Joby pivots successfully to a software-take-rate model on partner-operated aircraft + multiple expansion to aerospace/software-hybrid premium. By FY36 Joby operates 8,000+ aircraft globally, captures 16% of $250B UAM TAM, and software-take-rate revenue dominates with 35%+ mature margins.

Compound conditional: each piece individually plausible at 60-75%; joint at 3-5%. Reflected here at 3%. This is the genuine tail — not a 'mild upside,' but the scenario where Joby joins the aerospace/software giants club. Contributes $2.70 to the $7.81 weighted expected ($90 × 3%).

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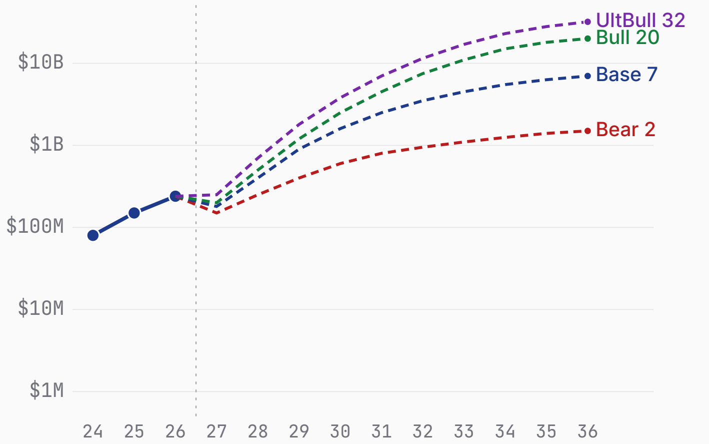
16 May 2026 · the snapshot

Joby Aviation business snapshot

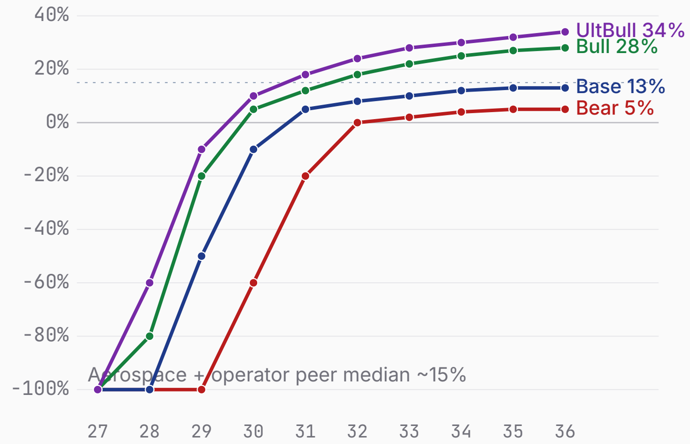
Bear $0.42 Base $3.65 Bull $30.05

FY24–FY26 history + FY27–FY36 scenario projections · fiscal years end Dec 31 · 10-K, Q1 2026 10-Q, Morgan Stanley UAM

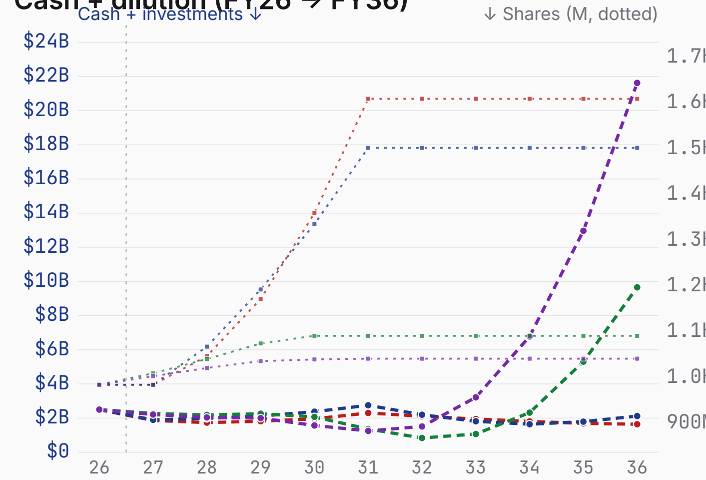
Revenue — history + scenarios to FY36 (log)



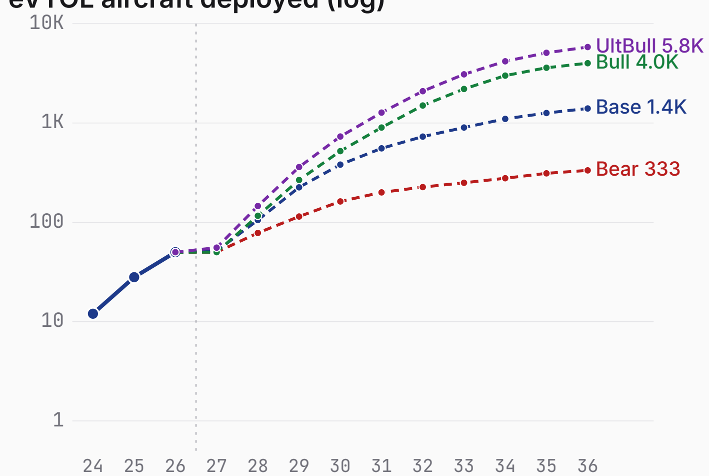
Path to profitability — op margin (FY27→FY36)



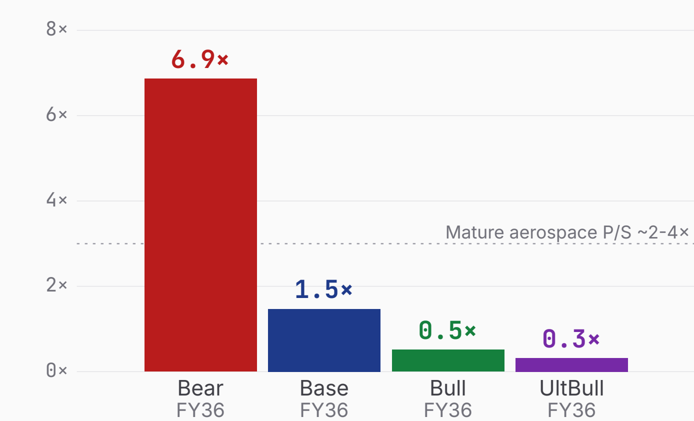
Cash + dilution (FY26 → FY36)



eVTOL aircraft deployed (log)

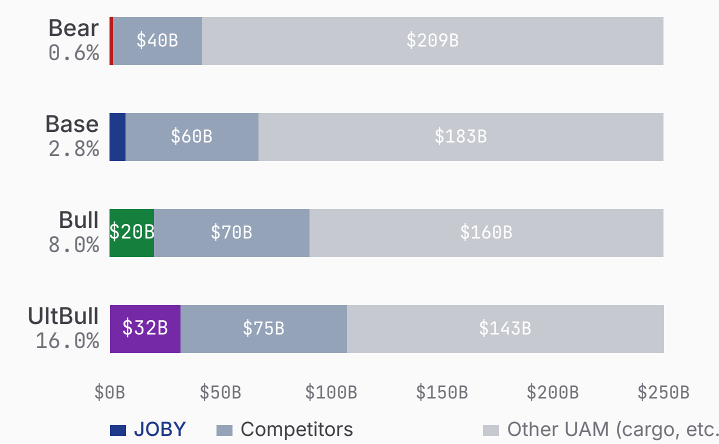


$10.30B mkt cap as P/S on FY36 rev



Bear=very stretched (6.9×), base=full (1.5×), bull=cheap (0.5×).

$250B global UAM TAM (FY36) — Joby share



Sources: JOBY 10-K FY25, Q1 2026 10-Q, Morgan Stanley UAM TAM Update (Global $1T by 2040; ~$250B FY36 interpolation).

INTERNAL RESEARCH · MEMO · NOT INVESTMENT ADVICE · AI-ASSISTED

16 May 2026 · the quantitative

Joby Aviation show your work

Bear $0.42 Base $3.65 Bull $30.05 · Weighted $8.44 (-19.6%)

BEAR $0.42

ASSUMPTIONS

TAM Year-10 ($B)	250
Mature share (%)	0.6
Mature op margin (%)	5
Sales-to-capital	0.8
Terminal WACC (%)	9.5
Terminal growth (%)	2.0
P(failure) (%)	35
Scenario probability (%)	30

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.15	-300%	-0.50	-0.44
FY28	0.25	-200%	-0.63	-0.48
FY29	0.40	-120%	-0.67	-0.46
FY30	0.60	-60%	-0.61	-0.37
FY31	0.80	-20%	-0.41	-0.22
FY32	0.95	+0%	-0.19	-0.09
FY33	1.10	+2%	-0.17	-0.07
FY34	1.25	+4%	-0.14	-0.05
FY35	1.40	+5%	-0.12	-0.04
FY36	1.50	+5%	-0.05	-0.02

Σ PV explicit FCF	-2.25
+ PV terminal (g 2.0%)	0.32

EQUITY BUILD · $B & PER SHARE

Operating EV	$-1.93B
+ Cash & investments	$2.50B
= Equity value	$0.57B
Raised over period	$2.75B
Dilution by FY36	+53%
FY36 shares (M)	1,508
DCF per share	$0.38
× (1−P_fail) [65%]	$0.25
+ P_fail × distress [35% × $0.50]	$0.17
= Expected per share	**$0.42**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$0.66	$1.04	$1.64	$2.58
0.06×	0.10×	0.16×	0.25×

BASE $3.65

ASSUMPTIONS

TAM Year-10 ($B)	250
Mature share (%)	2.8
Mature op margin (%)	13
Sales-to-capital	1.2
Terminal WACC (%)	8.5
Terminal growth (%)	2.5
P(failure) (%)	18
Scenario probability (%)	55

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.18	-250%	-0.51	-0.45
FY28	0.40	-120%	-0.66	-0.52
FY29	0.90	-50%	-0.87	-0.61
FY30	1.60	-10%	-0.74	-0.47
FY31	2.50	+5%	-0.63	-0.35
FY32	3.50	+8%	-0.61	-0.31
FY33	4.50	+10%	-0.48	-0.22
FY34	5.50	+12%	-0.31	-0.13
FY35	6.30	+13%	-0.02	-0.01
FY36	7.00	+13%	+0.14	+0.05

Σ PV explicit FCF	-3.03
+ PV terminal (g 2.5%)	6.10

EQUITY BUILD · $B & PER SHARE

Operating EV	$3.07B
+ Cash & investments	$2.50B
= Equity value	$5.57B
Raised over period	$3.75B
Dilution by FY36	+31%
FY36 shares (M)	1,284
DCF per share	$4.34
× (1−P_fail) [82%]	$3.56
+ P_fail × distress [18% × $0.50]	$0.09
= Expected per share	**$3.65**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$5.49	$8.25	$12.41	$18.66
0.52×	0.79×	1.18×	1.78×

BULL $30.05

ASSUMPTIONS

TAM Year-10 ($B)	250
Mature share (%)	8.0
Mature op margin (%)	28
Sales-to-capital	1.6
Terminal WACC (%)	7.5
Terminal growth (%)	3.0
P(failure) (%)	8
Scenario probability (%)	12

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.20	-200%	-0.46	-0.41
FY28	0.50	-80%	-0.59	-0.46
FY29	1.20	-20%	-0.68	-0.48
FY30	2.50	+5%	-0.69	-0.43
FY31	4.50	+12%	-0.71	-0.40
FY32	7.50	+18%	-0.53	-0.26
FY33	11.00	+22%	+0.23	+0.10
FY34	15.00	+25%	+1.25	+0.53
FY35	18.00	+27%	+2.98	+1.17
FY36	20.00	+28%	+4.35	+1.59

Σ PV explicit FCF	+0.94
+ PV terminal (g 3.0%)	32.10

EQUITY BUILD · $B & PER SHARE

Operating EV	$33.04B
+ Cash & investments	$2.50B
= Equity value	$35.54B
Raised over period	$2.05B
Dilution by FY36	+11%
FY36 shares (M)	1,091
DCF per share	$32.58
× (1−P_fail) [92%]	$29.97
+ P_fail × distress [8% × $1.00]	$0.08
= Expected per share	**$30.05**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$43.14	$61.93	$88.91	$127.65
4.11×	5.90×	8.47×	12.16×

ULTRA BULL $90.01

ASSUMPTIONS

TAM Year-10 ($B)	250
Mature share (%)	16.0
Mature op margin (%)	34
Sales-to-capital	1.8
Terminal WACC (%)	7.0
Terminal growth (%)	3.5
P(failure) (%)	4
Scenario probability (%)	3

10-YEAR DCF · $B

FY	Rev	Margin	FCF	PV FCF
FY27	0.25	-180%	-0.52	-0.47
FY28	0.70	-60%	-0.68	-0.55
FY29	1.80	-10%	-0.84	-0.62
FY30	3.80	+10%	-0.88	-0.58
FY31	7.00	+18%	-0.69	-0.42
FY32	11.50	+24%	+0.06	+0.03
FY33	17.00	+28%	+1.89	+0.93
FY34	23.00	+30%	+4.10	+1.85
FY35	28.00	+32%	+6.96	+2.88
FY36	32.00	+34%	+9.63	+3.67

Σ PV explicit FCF	+6.73
+ PV terminal (g 3.5%)	95.83

EQUITY BUILD · $B & PER SHARE

Operating EV	$102.56B
+ Cash & investments	$2.50B
= Equity value	$105.06B
Raised over period	$2.05B
Dilution by FY36	+11%
FY36 shares (M)	1,121
DCF per share	$93.72
× (1−P_fail) [96%]	$89.97
+ P_fail × distress [4% × $0.96]	$0.04
= Expected per share	**$90.01**

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
$126.24	$177.06	$248.34	$348.31
12.02×	16.86×	23.65×	33.17×

Joby Aviation supporting analysis · disclaimers · glossary

Bear $0.42 Base $3.65 Bull $30.05

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

1 **Type Cert is imminent.**
Stage 4 in progress, SR3 audit complete Q1'26. Conforming aircraft already flying. Removes the largest single bear risk.

2 **Best capitalized eVTOL.**
$2.5B cash exceeds Archer ($1.8B) and Vertical (+$850M raise). Multi-year runway through commercial scaling.

3 **Partnership stack is unique.**
Toyota ($894M, mfg), Delta ($200M, US distribution), Dubai RTA (exclusive 2032). No other eVTOL has this stack.

4 **Operator + mfg = wider moat.**
Joby owns/operates fleet rather than selling aircraft. Captures full revenue per flight, not just OEM margin.

5 **Mature margin may be conservative.**
Base case 13% reflects aerospace blend. If software-take-rate on partner aircraft scales, terminal margin could be 20%+.

FALSIFICATION TRIGGERS

Bear validation
Type Cert slips to 2027+ · Cash < $1.5B by EOY 2026 · Dubai passenger volume disappoints (< 50K rides FY27)

Bull validation
Type Cert achieved by Q3 2026 · Dayton facility hits 4 aircraft/month · Second exclusive country contract signed

Reframe needed
If Archer achieves Type Cert first — Joby's first-mover advantage in US erodes; share-of-TAM math compresses

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Scenarios, valuations, projections, and any forward-looking statements involve substantial assumptions and uncertainty. Actual results may differ materially from those projected. Past performance is not indicative of future results. The model is a model; reality is not.

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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

eVTOL — Electric Vertical Take-Off and Landing aircraft. Joby's S4 is a 5-
air-taxi rights in Dubai through 2032.

Type Certification — FAA approval to operate a new aircraft design
at Dayton, Ohio facility for production scale.

eIPP — eVTOL Integration Pilot Program. White House-backed initiative letting
existing routes (NYC-airport) for early ops.